

PROBABILITY THEORY & SYSTEMS

Pure Conceptual Mapping & Standardized Tutorial Question Matrix

Exclusions: Zero Historical Previous Year Papers (No PYQ Content)

Inclusions: Fundamental Axioms, Dependent Spaces, Continuous Integrals, and Higher Expected Moments

Standard: Higher Academic Education & Structural Engineering Frameworks — Proper English Copy

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CHAPTER 1: DEEP FOUNDATIONS OF PROBABILITY THEORY

1.1 The Mathematical Limit of Intuitive Spaces

In early mathematics, probability was defined simply as the count of successful outcomes divided by total possible variations. However, this structure breaks down completely when tracking infinite spaces or continuous intervals, where individual outcomes have a cardinality of zero. To overcome this limitation, advanced frameworks define probability as a normalized measure operating within a structural set algebra.

1.2 Kolmogorov's Core Postulates

Let S represent the complete sure sample field (the universe of all states) and A denote any valid sub-event within that universe. A mathematically sound probability measure P must satisfy three strict axioms:

I. The Postulate of BoundsThe probability of any event is a non-negative real measure:

$$P(A) \geq 0$$

II. The Postulate of CertaintyThe absolute probability of the entire sample universe is bounded at exactly one:

$$P(S) = 1$$

III. The Postulate of Countable AdditivityIf a collection of events $A_1, A_2, A_3, \dots$ are completely mutually exclusive (meaning no two events can occur simultaneously, or $A_i \cap A_j = \emptyset$ for all $i \neq j$), then the probability of their collective union is the exact sum of their individual probabilities:

$$P\left(\bigcup_{i=1}^{\infty} A_i\right) = \sum_{i=1}^{\infty} P(A_i)$$

1.3 Conditional Restructuring & Bayes' Network Matrix

When an external event B occurs, it acts as a constraint that shrinks the usable sample universe from S to B . The conditional probability of another event A occurring within this updated universe is defined as:

$$P(A | B) = \frac{P(A \cap B)}{P(B)} \quad \text{ext where } P(B) > 0$$

If a sample space is split into multiple mutually exclusive, exhaustive partitions ($B_1, B_2, \dots, B_n$), the total probability of any overlapping target event A is given by the **Theorem of Total Probability**. Reversing this relationship to update our initial assumptions based on new evidence leads directly to **Bayes' Theorem**.

$$P(B_k | A) = \frac{P(A | B_k) \cdot P(B_k)}{\sum_{i=1}^n P(A | B_i) \cdot P(B_i)}$$

1.4 Continuous Variable Matrices & Expected Densities

Continuous variables map smoothly across continuous scalar intervals. Since the probability of hitting an exact real number coordinate is zero ($\int_c^c f(x)dx = 0$), continuous systems use a **Probability Density Function (PDF)** denoted as $f(x)$.

The **Cumulative Distribution Function (CDF)** denoted as $F(x)$, measures the total accumulated probability from negative infinity up to a specific coordinate: $F(x) = P(X \leq x) = \int_{-\infty}^x f(t)dt$. By the rules of calculus, differentiating the cumulative distribution function recovers the original density function:

$$f(x) = \frac{d}{dx} [F(x)]$$

The center-of-mass balance coordinate of this distribution is its **Mathematical Expectation (Mean)** and its structural spread or dispersion is measured by its **Variance**

$$\mu = E[X] = \int_{-\infty}^{\infty} x \cdot f(x) \, dx \quad | \quad \text{Var}(X) = E[X^2] - (E[X])^2$$

CHAPTER 2: TEXTBOOK CORE QUESTION BANK

Every question in this chapter is designed to test your understanding of core concepts. They are entirely separate from previous exam question tracking paths.

Section 2.1: Axiomatic & Dependent Space Problems

Problem 1: Conceptual Proof of Complementary Spaces

Using Kolmogorov's axioms of probability, provide a formal mathematical proof showing that the probability of the empty set (the impossible event, denoted by $\emptyset$) is identically equal to zero ($P(\emptyset) = 0$).

Detailed Step-by-Step Solution:

1. Define a relationship between the sure sample space S and the empty set $\emptyset$. By the definition of set operations, the union of any set with the empty set leaves that set unchanged: $S \cup \emptyset = S$.
2. Additionally, a set and the empty set share no common elements, meaning they are completely disjoint or mutually exclusive: $S \cap \emptyset = \emptyset$.
3. Since these components are mutually exclusive, we can apply Kolmogorov's Third Axiom (Countable Additivity) to their union:

$$P(S \cup \emptyset) = P(S) + P(\emptyset)$$

4. Substitute the identity $S \cup \emptyset = S$ into the left side of the equation:

$$P(S) = P(S) + P(\emptyset)$$

5. According to Kolmogorov's Second Axiom, the probability of the entire sample universe is equal to unity ($P(S) = 1$). Substituting this value gives:

$$1 = 1 + P(\emptyset)$$

6. Subtracting 1 from both sides of the linear equation proves the theorem: $P(\emptyset) = 0$.

Problem 2: Probability Matrix Inversion (Bayes' Application)

An automated industrial sorting system uses three separate laser sensors (α , η , γ) to detect cracks in manufacturing components. The sensors inspect 40%, 35%, and 25% of the components, respectively. Their error rates (false positives) are known to be 2%, 3%, and 4%. If a component is flagged as defective, calculate the exact probability that it was inspected by sensor η .

Detailed Step-by-Step Solution:

1. Define the prior probabilities for each sensor: $P(\alpha) = 0.40$, $P(\eta) = 0.35$, and $P(\gamma) = 0.25$. Note that their sum is exactly $0.40 + 0.35 + 0.25 = 1.00$, forming a valid partition.
2. Define the conditional probabilities of a false positive error (E) for each sensor: $P(E|\alpha) = 0.02$, $P(E|\eta) = 0.03$, and $P(E|\gamma) = 0.04$.

3. Apply the Theorem of Total Probability to find the absolute system-wide error rate $P(E)$:

$$\begin{aligned}P(E) &= P(E|\alpha)P(\alpha) + P(E|\eta)P(\eta) + P(E|\gamma)P(\gamma) \\P(E) &= (0.02 \text{ imes } 0.40) + (0.03 \text{ imes } 0.35) + (0.04 \text{ imes } 0.25) \\P(E) &= 0.0080 + 0.0105 + 0.0100 = 0.0285\end{aligned}$$

4. Use Bayes' Theorem to find the updated posterior probability for sensor η given that an error occurred:

$$P(\eta | E) = \frac{P(E|\eta) \cdot P(\eta)}{P(E)} = \frac{0.03 \text{ imes } 0.35}{0.0285} = \frac{0.0105}{0.0285}$$

5. Simplifying the fraction yields the final precise answer: $P(\eta | E) = \frac{105}{285} = \frac{21}{57} = \frac{7}{19} \text{ approx } 0.3684$ (or 36.84%).

Problem 3: Normalization and Expectation of Continuous Spaces

A continuous random variable X has a probability density function defined by $f(x) = c(1 - x^2)$ within the interval $-1 \leq x \leq 1$, and $f(x) = 0$ outside this range.

- (i) Find the exact value of the normalization constant c .
- (ii) Calculate the mathematical expectation $E[X]$ of this distribution.

Detailed Step-by-Step Solution:

Part (i): Solving for the Normalization Constant c

1. To be a valid probability distribution, the total area under the density curve must equal 1: $\int_{-1}^1 c(1 - x^2)dx = 1$.

2. Pull out the constant and integrate the polynomial terms using standard calculus rules:

$$c \left[x - \frac{x^3}{3} \right]_{-1}^1 = 1$$

3. Evaluate the definite integral at the upper and lower limits:

$$c \left[\left(1 - \frac{1}{3} \right) - \left(-1 - \frac{-1}{3} \right) \right] = 1 \implies c \left[\frac{2}{3} - \left(-1 - \frac{1}{3} \right) \right] = 1$$

$$c \left[\frac{2}{3} - \left(-\frac{2}{3} \right) \right] = 1 \implies c \left[\frac{2}{3} + \frac{2}{3} \right] = 1 \implies c \left[\frac{4}{3} \right] = 1 \implies c = \frac{3}{4}$$

Part (ii): Calculating the Mathematical Expectation $E[X]$

1. Use the definition of mathematical expectation for continuous variables: $E[X] = \int_{-1}^1 x \cdot c \cdot f(x)dx$.

2. Substitute the normalized density function into the equation:

$$E[X] = \int_{-1}^1 x \cdot \left[\frac{3}{4}(1 - x^2) \right] dx = \frac{3}{4} \int_{-1}^1 (x - x^3)dx$$

3. Integrate the terms:

$$E[X] = \frac{3}{4} \left[\frac{x^2}{2} - \frac{x^4}{4} \right]_{-1}^1$$

4. Evaluate the definitive boundaries:

$$E[X] = \frac{3}{4} \left[\left(\frac{1}{2} - \frac{1}{4} \right) - \left(\frac{1}{2} - \frac{1}{4} \right) \right] = \frac{3}{4} \left[\frac{1}{4} - \frac{1}{4} \right] = 0$$

Conceptual Insight: The mean value $E[X] = 0$ makes complete physical sense because the density function $f(x) = \frac{3}{4}(1 - x^2)$ is perfectly symmetric around the vertical y-axis over the interval $[-1, 1]$, meaning its center of mass lies exactly at the origin.